MIL-C-22585A(AS)
25 May 1984
SUPERSEDING
MIL-C-22585(WEPS)
12 July 1960

MILITARY SPECIFICATION

COMPUTER, PRESSURE REDUCTION, METEOROLOGICAL CP-402/UM( )

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a pressure reduction computer for use at weather stations located at various elevations to compute sea-level pressure and altimeter setting from station barometric pressure readings.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-406	Plastic, Organic, General Specification, Test Methods
L-P-535	Plastic Sheet(Sheathing and Film); Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid
QQ-B-626	Brass Leaded and Nonleaded; Rod, Shaped, Forgings, and Flat Products with Finished Edges(Bar and Strip)
QQ-N-290	Nickel Plating(Electrodeposited)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Naval Air Engineering Center, ESSD, Engineering Specifications and Standards Department, (Code 93), Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6660

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SPECIFICATIONS (Continued)

Federal (Continued)

QQ-P-330	Phosphor Bronze Bars, Plates, Rods, Sheets, Strips, Flatwire, and Structural and Special Shaped Sections

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attribute
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-794	Parts and Equipment, Aeronautical, Preparation for Delivery

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the computer furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified in 4.4 through 4.4.3 (see 6.2).

3.2 Materials. All plastic parts of the computer shall be fabricated of vinyl chloride polymer and copolymer rigid sheets, conforming to L-P-535, and shall be of Class 1 transparent, or Class 3 opaque white, as required. After printing, the printed material shall be sealed with a protective lamination of Class 1 transparent

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vinyl plastic. All sharp edges shall be removed by polishing. All laminations shall be free of peels and blisters. All metal parts of the center pivot assembly shall be of brass, Composition 22, half hard, per QQ-B-626, except for the spring washers which shall be of phosphor bronze, Composition A, spring hard, per QQ-P-330, and shall be nickel plated Class 1, bright finish, Type VI (KC) per QQ-N-290. All metal parts shall be free from burrs.

3.3 Design and construction.

3.3.1 Scale layouts shall be photographic reproductions of master to be furnished on loan as set forth in the terms of the contract or order, except for such deletions and insertions of title legends as provided for in paragraph 3.5.

3.3.2 Component parts. The Pressure Reduction Computer CP-402/UM consists of the following components:

a. Base Disc

b. Rotor Disc (Sea-Level Reduction)

c. Rotor Disc (Altimeter Setting)

d. Cursor

e. Pivot Assembly

3.3.3 Base disc. The base disc shall contain P, Po, and the P scales and centering check marks on the "Sea-Level Reduction" face, and the P, A. S., and the P scales, and centering check marks on the "altimeter setting face." See Figures 1 and 2. The disc shall be fabricated of Class 3 opaque white vinyl plastic 0.070 ± 0.005 inch thick and shall have a diameter of 11.00 ± 0.03 inches. It shall be securely bonded after printing with laminations of 0.005 inch clear transparent vinyl plastic on each side, and shall be free of warp and wind.

3.3.4 Rotor disc (sea-level reduction). The "Sea-Level Reduction" rotor disc shall contain the r scales, P index, printed text entitled "Scale Symbols" and "Computation of Pressure Reduced to Sea-Level," Centering Check marks. See Figure 1. The disc shall be constructed of Class 1 transparent vinyl plastic 0.040 ± 0.005 inch thick, and shall have a diameter of 10.50 ± 0.03 inches. Scales and text shall be printed in reverse on the under side of the rotor disc in order to reduce parallax. Subsequently, an opaque pastel green backing shall be applied to (a) a circular area extending from the center of the disc to a radius of 7.70 centimeters, and (b) an annulus extending between the radii of 9.80 and 11.50 centimeters. After application of the opaque backing, a protective

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lamination of 0.005 inch clear, transparent vinyl plastic shall be securely bonded to the printed side of the disc. Embossed spots shall be spaced at 22½ degree intervals around the disc and shall be of sufficient depth to maintain 0.010 inch clearance between the under side of the rotor disc and the base disc. The spots shall be placed at a radius of 11.5 centimeters. The disc shall be free of warp and wind.

3.3.5 Rotor disc (Altimeter Setting). The "Altimeter Setting" rotor disc shall contain the H scale, P index, printed text entitled "Scale Symbols", "Altimeter Setting Computation", and "Pressure-Altitude Computation", centering check marks. See Figure 2. The disc shall be constructed of Class 1 transparent vinyl plastic 0.040 ± 0.005 inch thick, and shall have a diameter of 10.50 ± 0.03 inches. Scale and index shall be printed in reverse on the under side of the rotor disc in order to reduce parallax.

Subsequently, an opaque yellow backing shall be applied to (a) a circular area extending from the center of the disc to a radius of 10.30 centimeters, and (b) an annulus extending between radii of 11.2 and 12.30 centimeters. After application of the opaque backing a protective lamination of 0.005 inch clear, transparent vinyl plastic shall be securely bonded to the printed side of the disc. Embossed spots shall be spaced at 22-½ degree intervals around the disc and shall be of sufficient depth to maintain 0.010 inch clearance between the under side of the rotor disc and the base disc. The spots shall be placed at a radius of 11.2 centimeters. The disc shall be free of warp and wind.

3.3.6 Cursor. The cursor shall be fabricated in accordance with Figure 3, a part of this specification. The hairline on the cursor plate shall be on the side of the plate nearest the rotor disc and beneath the 0.005 inch lamination. The cursor shall be mounted on the pivot assembly with such precision that the unit can be set to any point of the base scale without binding on the base or rotor discs. Moreover, the tension of the pivot assembly spring washers shall be such that a torque of approximately 25 oz-in. is required to move the cursor from its rest position at any point on the base disc scales.

3.4 Scales, text and printing.

3.4.1 Scales and text. Reduced reproductions of the scales, centering check marks, and text, required on the Sea-Level Reduction and Altimeter Setting sides of the computer, are shown respectively in Figure 1 and 2. Reproductions of the lettering and hairlines required on the two arms of the cursor are also shown in Figures 1 and 2. The procuring activity will supply the contractor with either regular or reverse plastic film negatives or positives, as required by him in his printing process, from which plates can be prepared for printing the scales, centering check marks, cursor lettering, text, and opaque backings, required in fabrication of the computer. The film copies supplied shall remain the property of the Government, and shall be returned to the procuring activity at the completion of the contract.

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3.4.2 Printing. All material reproduced on the discs shall be printed in dense opaque black. Each line, figure, and letter, shall be sharp and cleancut, and of uniform width. There shall be no appreciable distortion resulting from the printing or laminating processes. Hairline and scale symbols appearing on the cursor arms shall be printed in dense, non-fading red ink. Scales shall be free from distortion, line-breaks, or imperfections, and shall be printed so that the angular position of each graduation shall agree within 0.005 inch of its position on the original film negatives (or positives) supplied to the contractor. The line width of graduation lines on the printed computers and the hairline on the cursor shall fall within the following ranges: minimum = line-width on negative supplied to contractor minus 0.0015 inch; maximum = line-width on negative supplied to contractor plus 0.003 inch.

3.5 Identification marking. The following data shall be inscribed on the two rotor discs at the location shown in figures 1 and 2:

a. Procuring activity

b. Stock No. (to be assigned)

c. Contract No. (to be assigned)

d. Manufacturer's Name, City and State

3.5.1 Nomenclature. Nomenclature assignments and approval of identification markings shall be obtained in accordance with MIL-N-18307.

3.6 Workmanship. Workmanship shall be first class in all respects and in accordance with the best commercial practice. All component parts and scales shall be free from any defects which may adversely affect the appearance and serviceability of the computer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the computer shall be classified as follows:

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a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.1 and 4.3 through 4.3.3).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4).

4.3 First article inspections. The first article inspections of the computer shall consist of all the examinations and tests set forth in this specification.

4.3.1 First article inspection instructions. The first article inspection samples shall consist of three computers, of each manufacturer's part number to be furnished in accordance with the contract or order. The manufacturer shall submit certified inspection results showing conformance with all the material requirements and tests of this specification. The samples and certified inspection report shall be submitted to the testing activity designated in the contract or order.

4.4 Government inspection. The right is reserved for government inspectors to conduct inspection of equipment after it has been submitted for acceptance on the contract. Such inspection being in addition to that performed by the contractor (4.2). In order to facilitate government inspection, the contractor shall submit the equipment in discrete lots if so required by the contracting officer. At the discretion of the Government inspector, inspection shall be conducted by either of the following methods or a combination of both:

a. 100-percent inspection.

b. Sampling inspection. Samplings and rejection procedures shall be in accordance with MIL-STD-105. The acceptable quality level (AQL) for major defects shall be 1% and 4% for minor defects. The acceptance of the lot shall be contingent upon results of the government inspection.

4.4.1 Disposition of non-conforming products. When defective sample units or rejected lots are resubmitted for acceptance such items shall be suitably tagged to indicate cause of failure and means employed to correct the fault.

4.4.2 Unspecified tests. Government personnel may perform such tests as are considered necessary to determine compliance with the specifications even though the tests are not required by the specification. Such tests may include any of those contained in L-P-406 and will normally be performed at a government laboratory or station on first article or

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production samples. Costs of equipment tested and expense of testing under this paragraph shall be borne by the government. The right is reserved for government inspectors to withdraw materials, parts or equipment from production and conduct non-destructive tests after which the items will be returned to production. The contractor should provide such testing equipment as may be necessary if the equipment is available.

4.5 Visual and mechanical inspection. All units on order shall be given a visual and mechanical inspection to determine compliance with the requirements contained herein. To insure that the rotor discs are centered and mounted to the required tolerances, the computer will be inspected for eccentricity, employing the centering check marks provided on the base and rotor discs. In performing this inspection each of the four marks on the rotor discs will be placed over the long centering mark of one of the base disc centering groups. The other three centering marks on the rotor disc shall then lie between the outside centering marks of each set on the base disc. Each of the four centering marks on the rotor discs shall, in turn, be placed over the same set of marks on the base disc and the remaining three marks checked as described. If at any of these settings, one or more of the marks on the rotor disc falls outside the outer of any set of three marks on the base disc, the computer will be rejected. The use of this system of check marks will restrict allowable eccentricity errors at the periphery of the rotor discs to 0.015 inch or less.

5. PACKAGING

5.1 Preservation, packaging and packing. The computers shall be packaged in accordance with MIL-STD-794. Preservation for level A shall be in accordance with MIL-P-116, Method IA.

5.2 Marking for shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows: Computer, Pressure Reduction, Meteorological CP-402/UM( ).

6. NOTES

6.1 Use. The computers described in this specification are intended for use by weather personnel to compute sea-level pressure and altimeter setting from station barometric pressure readings.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of specification.

b. Level of packaging and packing required.

c. Quantity of computers desired.

d. Whether first article is required (see 3.1).

e. Point of first article testing.

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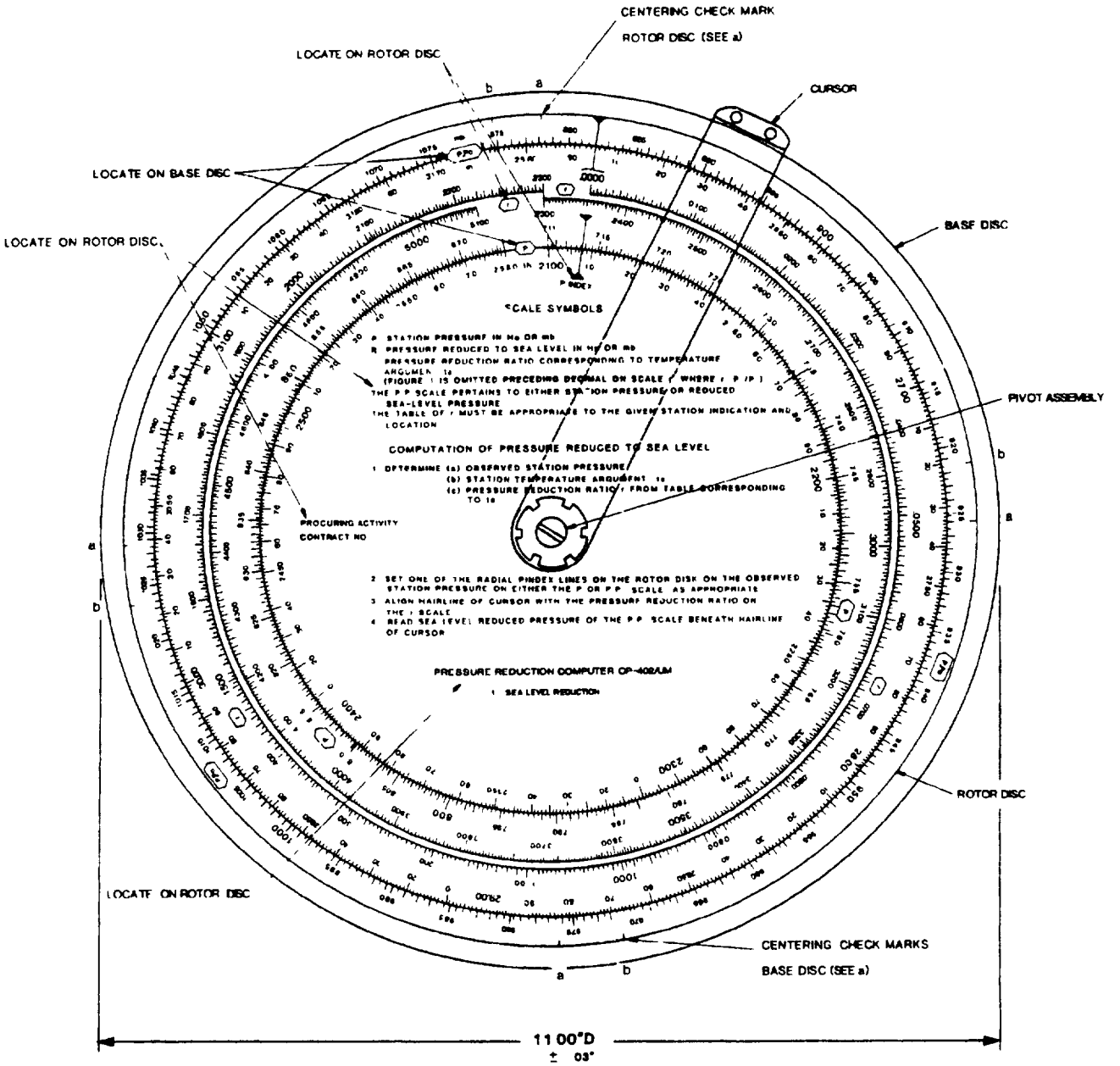


FIGURE 1.

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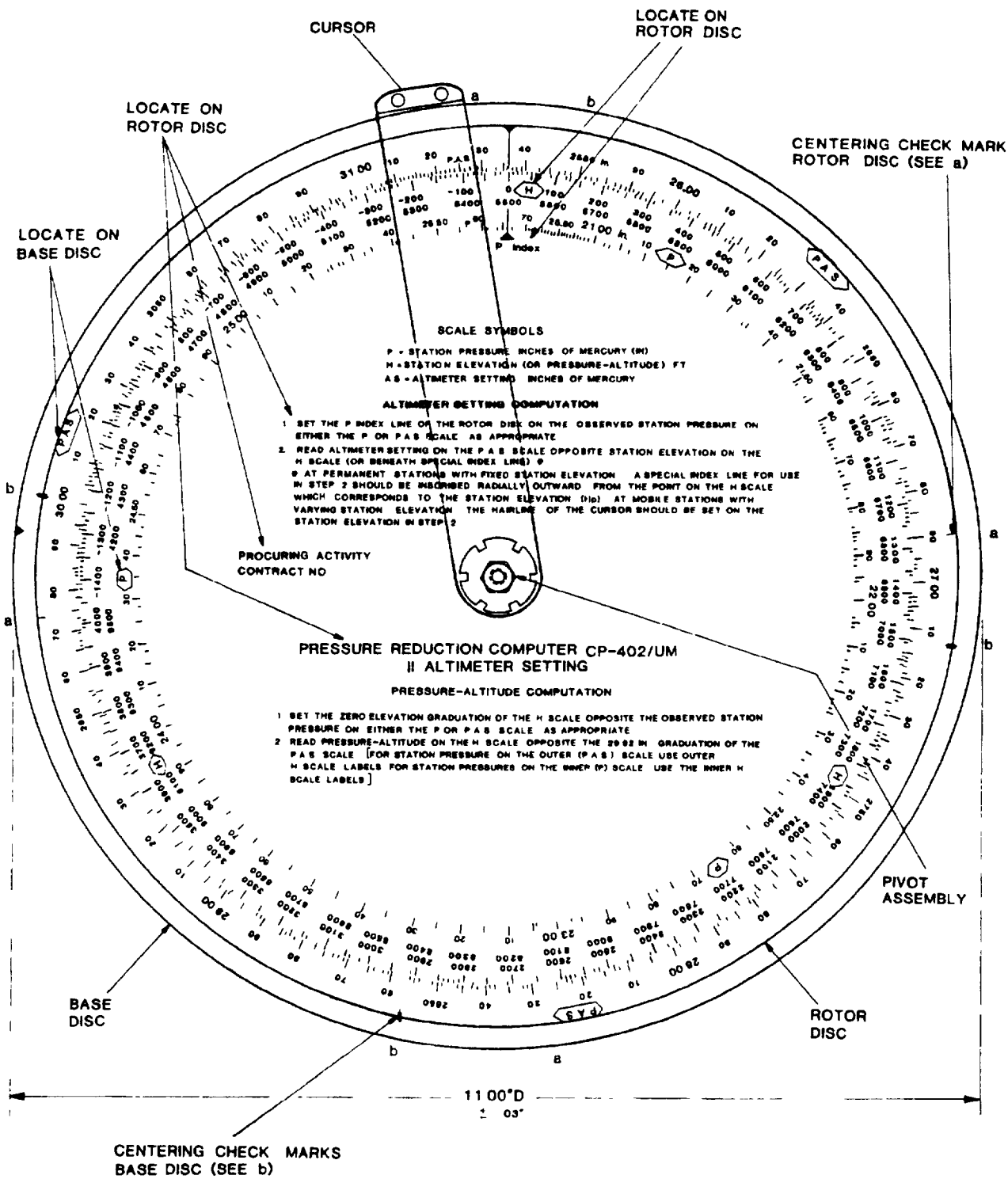
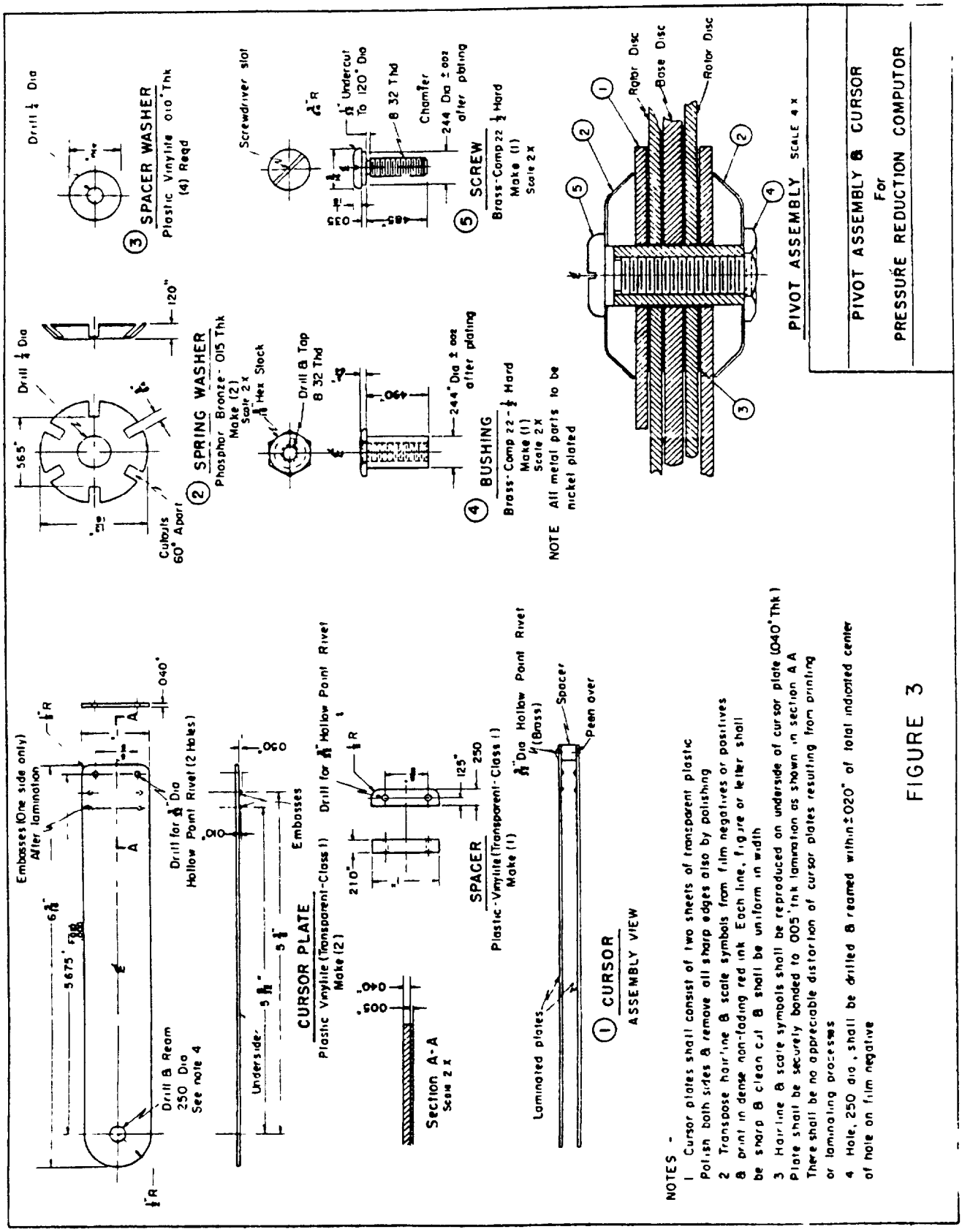


FIGURE 2.

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NOTES -

1. Cursor plates shall consist of two sheets of transparent plastic Polish both sides & remove all sharp edges also by polishing
2. Transpose hairline & scale symbols from film negatives or positives & print in dense non-fading red ink. Each line, figure or letter shall be sharp & clean cut & shall be uniform in width
3. Hairline & scale symbols shall be reproduced on underside of cursor plate (.040" Thk) Plate shall be securely bonded to .005" thk lamination as shown in section A A There shall be no appreciable distortion of cursor plates resulting from printing or laminating processes
4. Hole, .250 dia., shall be drilled & reamed within ±.020" of total indicated center of hole on film negative

FIGURE 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-C-22585A(AS)	COMPUTER, PRESSURE REDUCTION, METEOROLOGICAL CP-402/UM( )

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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